

Skills Summary

Ratios and Equivalent Ratio Tables

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a ratio off a model (ratio-from-model)

Three ways to write the same ratio:

$$a \text{ to } b \quad a : b \quad \frac{a}{b}$$

Order is part of the meaning. The quantity named *first* in the question is written first. Count the model, then simplify by dividing both numbers by a common factor — the model does not change, only the notation.

Example: A strip shows 6 blue counters and 9 white counters. Write blue to white in simplest form.

Step 1. The question names blue first, so blue is the first number: this is a part-to-part ratio read straight off the model.

Step 2. Blue to white is 6 to 9, or $6 : 9$, or $\frac{6}{9}$; dividing both by 3 gives $\frac{2}{3}$ — for every 2 blue there are 3 white.

Try it: A strip shows 10 red counters and 4 green counters. Write red to green in simplest form.

check: $\frac{5}{2}$

Skill 2 — Part-to-part vs. part-to-whole (part-to-whole)

Rule:

If the ratio of two parts is $a : b$, the whole is $a + b$ shares.

$$\text{part to part} = \frac{a}{b} \quad \text{part to whole} = \frac{a}{a + b}$$

Only the **part-to-whole** form may be read as “that fraction of the group.”

Example: A fruit bowl has apples and oranges in the ratio 3 to 5. What fraction of the fruit is apples?

Step 1. The question asks for a fraction *of the whole bowl*, so build the whole first instead of using the 5.

Step 2. The whole is $3 + 5 = 8$ shares and apples are 3 of them, so the fraction is $\frac{3}{8}$.

Try it: A team has forwards and defenders in the ratio 4 to 6. What fraction of the team is forwards?

check: $\frac{2}{5}$

Skill 3 — Equivalent ratio tables (equivalent-ratio-table)

Rule:

Multiply (or divide) **both** quantities by the same number to get an equivalent ratio. Every column of a ratio table names the same relationship.

Setting two columns equal gives a proportion: $\frac{x}{\text{new}} = \frac{\text{old part}}{\text{old whole step}}$, then cross-multiply.

Example: 4 tickets cost 10 dollars. What do 6 tickets cost?

Step 1. Cost per ticket is steady, so the two columns are equivalent ratios — write a proportion rather than adding.

Step 2. $\frac{x}{6} = \frac{10}{4}$, so $4x = 60$ and the cost is **15** dollars. (Check: 2 tickets cost 5 dollars, and $3 \times 5 = 15$.)

Try it: 3 books cost 12 dollars. What do 7 books cost?

check: 27

Skill 4 — The value of a ratio (value-of-ratio)

Rule:

The **value** of the ratio $a : b$ is the quotient $a \div b$ — one number, the unit rate “ a -units per 1 b -unit.”

Two ratios are equivalent exactly when their values are equal, so comparing values is how you decide which of two ratios is larger.

Example: A car travels 18 miles in 3 hours. Write the ratio of miles to hours and find its value.

Step 1. “Per hour” means the hours become 1, so divide the first quantity by the second.

Step 2. The ratio is $\frac{18}{3}$, and its value is $18 \div 3 = 6$ miles per hour.

Try it: A typist types 45 words in 5 minutes. Find the value of the ratio of words to minutes.

check: 6

(!) **Watch out:** $3 : 5$ and $\frac{3}{5}$ look alike but answer different questions when the parts are *both* parts: $3 : 5$ says 3 reds for every 5 blues, while “ $\frac{3}{5}$ of the group” would need the whole, which is 8. Ask “compared with the other part, or with the whole?” before writing.